

## Dokumentasi Alur Data & Rumus Bisnis

Mill Onboarding · Supplier Due Diligence · Modul Terkait

Sumber valid dari codebase (main.js, GoogleAppsScript-backend-v3-full.gs)

# Daftar Isi

Ringkasan 15 bagian

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# 1. Arsitektur Sistem

Frontend → API Proxy → Google Apps Script → Google Sheets

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- Frontend: Vite SPA (src/main.js + modul panel per fitur)
- API: /api/gas-proxy → doGet / doPost di Google Apps Script
- Database: Google Sheets — setiap modul = satu atau lebih tab sheet
- Backend canonical: scripts/GoogleAppsScript-backend-v3-full.gs
- Prinsip penting:
  - Skor dan risk level mill dihitung di Google Sheet (formula), bukan di web app
  - Web app tidak menulis ulang kolom formula saat save

```
[Supplier DD] [Mill Onboarding] [TTM/TTP] [EUDR] [Monthly Report]
↓
/api/gas-proxy
↓
Google Apps Script
↓
SDD_MAIN · Mill Profile · Monitoring TTP/TTM · ...
```

## 2. Supplier Due Diligence (SDD)

Panel: #panel-supplier-dd

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- Penyimpanan: SDD\_MAIN (1 baris/submission), SDD\_MILL\_LIST (Traceability A), SDD\_FFB\_LIST (Traceability B)
- Primary key: submission\_id (format SUB-YYYYMMDD-XXXXXX)
- Supplier type: MILL, KCP, atau TRADER
- Entry point:
  - Upload Excel — Main Form + Traceability A/B
  - Screening form manual — SCR fields, GRV, PRI
  - Approve / Hold / Reject oleh approver
  - NBL Check vs sheet NBL & Unilever NBL
- Legacy: sheet SDD Data (flat) masih didukung via adapter

### 3. SDD — State Machine & Alur

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Import Excel
↓ Save Draft → SCR Status: Draft
↓ Submit → SCR Status: Submitted
↓ Boss Decision:
Approve → statusSDD = APPROVED
Hold → statusSDD = ON HOLD
Reject → statusSDD = REJECTED
↓ (APPROVED + Submitted) → Mill Onboarding Task List
↓ '+ Add to Mill' → Mill Onboarding Profile
↓ mill_ttp_sync → Monitoring TTP/TTM
↓ mill_added = true → hilang dari Task List
```

**Catatan:** TTP tidak di-sync saat SDD di-approve saja. Sync TTP terjadi setelah Mill disimpan dari Task List dengan payload mill\_ttp\_sync.

## 4. SDD → Kemana Data Mengalir?

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| Trigger                  | Target Modul            | Fungsi / Mekanisme                   |
|--------------------------|-------------------------|--------------------------------------|
| statusSDD = APPROVED     | Contact List Supplier   | syncContactFromApprovedSubmission_() |
| APPROVED + Submitted     | Mill Task List          | isApprovedSubmittedSddRow_()         |
| Save Mill dari Task List | Mill Onboarding Profile | add / update sheet mill              |
| mill_ttp_sync            | Monitoring TTP/TTM      | syncTtpFromMillOnboarding_*          |
| NBL Check                | SCR - No Buy List       | runSddNblCheck()                     |
| Monthly Report           | Section 01 SDD          | fetchSddList()                       |
| Export PDF               | SDD Screening PDF       | sddExportPdf()                       |

## 5. Mapping Field: SDD → Mill Onboarding

Fungsi: mapSddRowToMillPayload() / mapSddTmlRowToMillPayload()

| Kolom Mill               | Sumber SDD                             |
|--------------------------|----------------------------------------|
| GROUP NAME               | Group Name                             |
| COMPANY NAME             | Company Name                           |
| MILL NAME                | Mill Name / KCP Name / TML - Mill Name |
| COORDINATES              | Latitude + Longitude (comma-separated) |
| MILL CAPACITY (TON/HOUR) | Mill Capacity (Ton/Hour)               |
| MILL CATEGORY            | Mill Category / KCP Category           |
| ADDRESS                  | Mill Address / KCP Address             |
| MONTH, YEAR              | Periode SDD                            |
| SOURCE TYPE              | MILL / KCP / TRADER                    |
| TRADER NAME              | Nama trader parent (jika applicable)   |
| UML ID                   | TML - UML ID (TML lines)               |

## 6. Mapping: SDD FFB → Monitoring TTP/TTM

Prinsip: 1 baris TTP = 1 pemasok FFB (Traceability B) + konteks Main Form

| Kolom TTP                      | Sumber SDD                   |
|--------------------------------|------------------------------|
| GROUP / COMPANY / MILL NAME    | Mill Onboarding identity     |
| FFB SUPPLIER GROUP / NAME      | FFB - Supplier Group/Name    |
| CATEGORY                       | FFB - Supplier Category      |
| LAT, LONG                      | FFB - Latitude/Longitude     |
| VILLAGE, SUBDISTRICT, DISTRICT | FFB location fields          |
| CONCESION / PLANTED AREA       | FFB area fields              |
| LEGALITAS, ISPO/RSPO/ISCC      | FFB certification            |
| FFB SUPPLY to MILL (TON)       | FFB - Total Supply FFB (Ton) |
| submission_id, ffb_line_id     | Metadata sync (upsert)       |

Referensi lengkap: docs/sdd-to-ttm-ttp-field-mapping.md



## 7. Mill Onboarding — Entry & Output

Sheet: Mill Onboarding Profile (CPO/PK) · Mill Onboarding Waste (POME/SHELL)

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### Cara data masuk:

- Manual Add/Edit — modal #btn-add-mill
- SDD Task List — '+ Add to Mill' dari approved screening
- Supply Excel Import — upload → draft → submit

### Supply routing:

- CPO/PK → sheet mill
- POME (ISCC/INS) / SHELL → sheet millWaste
- KCP plant → FACILITY NAME PK
- Refinery plant → FACILITY NAME CPO

### Data mengalir ke:

| Consumer                    | Join Key                   | Data Dipakai            |
|-----------------------------|----------------------------|-------------------------|
| Monitoring TTP/TTM          | GROUP COMPANY MILL<br> UML | Identity + FFB sync     |
| Monthly Report \$02         | Period + risk level        | High-risk MILL rows     |
| Performa Facility           | FACILITY NAME<br>CPO/PK    | Supply, traceability %  |
| Questionnaire<br>Monitoring | group company mill         | Registry mill           |
| EUDR Potential              | group company mill         | Identity sync           |
| Mill Executive Report       | Quarter                    | Supply, risk, NBL, cert |

## 8. Supply Task List — Alur

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Upload Excel (PLANT, CATEGORY, COMPANY, MILL, QTY)

↓ Parse & match profil mill

↓ match\_status: matched | new | group\_mismatch

↓ Save Draft → Supply Import Draft

↓ Review di Task List modal

↓ Submit → submitSupplyDraft\_

↓ Append row → Mill Profile / Mill Waste

↓ Google Sheet hitung kolom formula

- CPO/PK submit: kolom produk lain dikosongkan (CPO submit clear PK, dst.)
- Formula columns di-restore dari baris di atas — tidak di-overwrite
- Repair drift: reconcileSupplyDraft\_()

## 9. Rumus di Google Sheet (Read-Only dari Web)

Kolom MILL\_FORMULA\_HEADERS\_ — tidak pernah ditulis dari web app

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- Skor: SCORE, DEFORESTATION/BURN/PEAT SCORE, TOTAL SCORE SPATIAL, TOTAL SCORE, LEGALITY SCORE
- Agregat: TOTAL GRIEVANCES, TOTAL POLICY, TOTAL CERTIFICATION
- Risk: RISK LEVEL, RESULT RISK LEVEL, BUYER NO BUY LIST, PRIORITY ENGAGEMENT, SUPPLIER STATUS
- Supply %: PERCENTAGE SUPPLY CPO/PK/ISCC/INS/SHELL, PRODUCT SUPPLY
- Waste: TOTAL POME SUPPLY, MAX SUPPLY POME/SHELL, REMAINING STOCK, TOTAL SCORE SUPPLY
- Status: COMPLIMENT/NOT COMPLIMENT, DECLARATION MONITORING, CPO, PK
- SD Monitoring: LAST SD DATE, DAY LEFT, Status, Result, Risk Number

## 10. Rumus App — Grievance, TTM/TTP, EUDR

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### Grievance Risk Score (src/grievance-risk.js)

- Total Score = jumlah 6 indikator (skor 1–3 masing-masing)
- Total  $\leq 8 \rightarrow$  Low · Total  $\leq 11 \rightarrow$  Medium · Total  $> 11 \rightarrow$  High
- Indikator: Publish, Subject, Repeat, Consequence, Group Scale, No Response

### TTM Coordinate % (ttpCalcTtmCoordinatePct\_)

- Pool: SOURCE TYPE MILL/TRADER/REFINERY dan SUPPLY CPO/PK  $> 0$
- Traceable: baris dengan koordinat valid
- $\% = \Sigma(\text{qty traceable}) / \Sigma(\text{qty total}) \times 100$

### TTP Aggregate % (ttpAggregateTotalTraceablePct\_)

- Primary:  $\Sigma(\text{Traceable Volume}) / \Sigma(\text{CPO/PK SUPPLY}) \times 100$
- Fallback: rata-rata kolom % CPO/PK TRACEABLE

### EUDR Potential Status (eudrComputeStatus\_)

- Semua kriteria ENABLED harus PASS  $\rightarrow$  Potential
- Salah satu FAIL  $\rightarrow$  Not Potential
- Kriteria: legality, millCategory (Integrated), ownPlasmaFfb ( $\geq 70\%$ ), resultRiskLevel (Low),
  - millLocation (APL), certification, grievance, ndpePolicy, noBuyList, deforestation

## 11. Mill Risk Reason Pills

src/mill-risk-reason.js — interpretasi gap field untuk UI

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- No Coordinate — koordinat kosong/invalid
- Legality Not Complete — LEGALITY SCORE  $\neq$  1
- Non APL Area — MILL LOC bukan APL
- No NDPE / No Certification
- Deforestation — RISK REDUCTION FACTOR = 1 atau width > 0 ha
- High Deforestation — factor = 2 atau width > 25 ha
- On No Buy List — BUYER NO BUY LIST = Yes
- Mills dengan RESULT RISK LEVEL = Low tidak menampilkan reason pills

## 12. Peta Modul & Koneksi

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SDD (APPROVED) ■■→ Contact List  
SDD ■■→ Mill Task List ■■→ Mill Profile ■■→ TTP sync  
Supply Excel ■■→ Mill Profile  
Mill ■■→ EUDR Potential TTP FFB ■■→ EUDR FFB %  
Mill + TTP + Grievance + NBL + EUDR ■■→ Monthly Report  
Mill + TTP + Supplied CPO/PK ■■→ Performa Facility  
NBL ■■→ SDD screening + Mill BUYER NO BUY LIST (sheet formula)

## 13. Daftar Google Sheets (Database)

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| Key             | Tab Name                | Modul                  |
|-----------------|-------------------------|------------------------|
| mill            | Mill Onboarding Profile | Mill Onboarding        |
| millWaste       | Mill Onboarding Waste   | Mill Waste             |
| supplyDraft     | Supply Import Draft     | Supply Task List       |
| sddMain         | SDD_MAIN                | Supplier Due Diligence |
| sddMill         | SDD_MILL_LIST           | SDD Traceability A     |
| sddFfb          | SDD_FFB_LIST            | SDD Traceability B     |
| ttp             | Monitoring TTP/TTM      | TTM/TTP                |
| grievance       | Grievance Monitoring    | Grievance              |
| nbl             | NBL / Unilever NBL      | No Buy List            |
| eudrPotential   | EUDR Potential          | EUDR Potential         |
| eudrDds         | EUDR DDS (+ child)      | EUDR DDS export EU     |
| contactSupplier | Contact List Supplier   | Contact List           |

## 14. Perbedaan: SDD vs EUDR DDS

| Aspek   | SDD                             | EUDR DDS                       |
|---------|---------------------------------|--------------------------------|
| Panel   | #panel-supplier-dd              | #panel-due-diligence-statement |
| Storage | SDD_MAIN / MILL / FFB           | EUDR DDS + child sheets        |
| Tujuan  | Screening internal + onboarding | Paket export EU EUDR           |
| Output  | Task List → Mill → TTP          | DOCX/PDF untuk buyer           |
| Code    | src/main.js                     | src/dds-ui.js                  |

Keduanya terpisah — tidak ada coupling langsung di codebase.



## 15. Aturan Validasi & Guard Rules

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- SDD Submitted tidak bisa downgrade ke Draft
  - Task List: SCR Status = submitted AND statusSDD = APPROVED
  - TTP sync: APPROVED + submitted + identity lengkap (GROUP/COMPANY/MILL/UML)
  - Supply submit: match\_status = matched atau new; company wajib
  - Mill save: kolom formula di-strip sebelum write
  - EUDR: semua kriteria enabled harus pass untuk Potential
  - Grievance: 6 indikator wajib diisi sebelum klasifikasi
  - Koordinat: normalizeCoordinate() + recoverCoord() untuk locale Indonesia
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*Dokumentasi valid dari codebase · Tidak ada perubahan kode aplikasi*